

E-Commerce Sales Performance Analytics

Data Analytics Internship Project Report

Project Objective: Analyze e-commerce transaction data, calculate business KPIs, identify trends and performance drivers, and translate findings into actionable recommendations.

1. Project Overview

The project follows a practical analytics workflow: data generation and preparation, data-quality checks, KPI calculation, aggregation by time/category/region/channel, visualization, interpretation, and business recommendations.

The dataset is synthetic and reproducible. It contains 1,200 transactions for 2025 and is generated inside the Python program using a fixed random seed.

2. Technology Stack

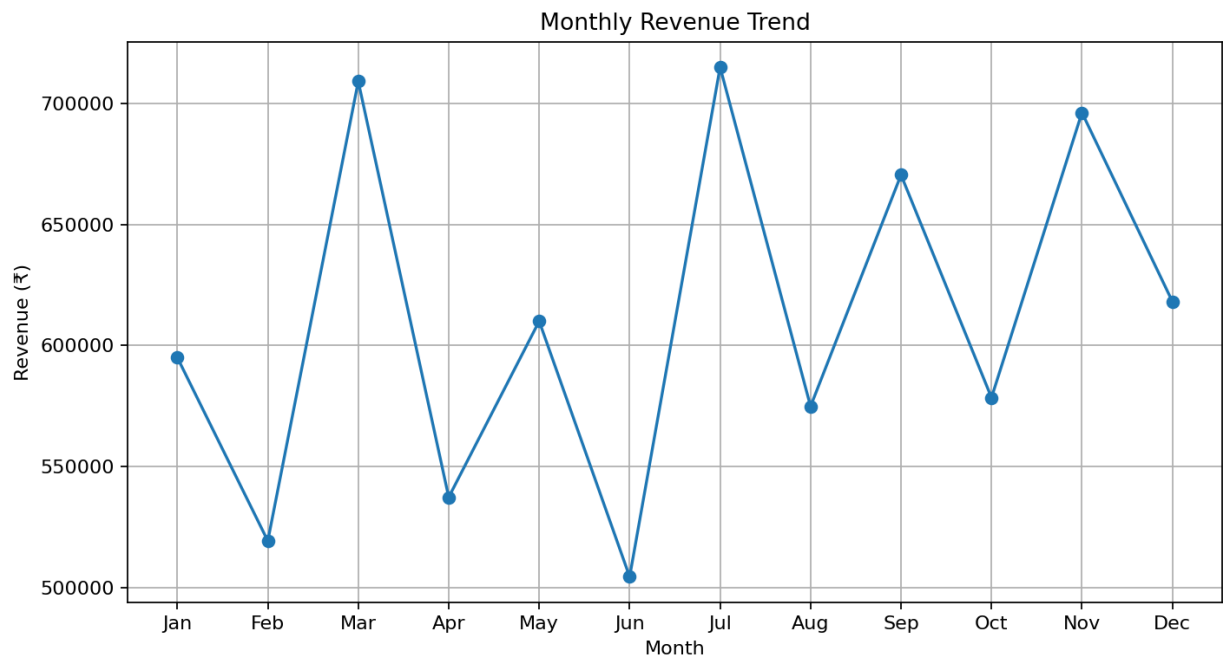
Python, Pandas, NumPy and Matplotlib. The code can be executed in VS Code, Jupyter Notebook or Google Colab.

3. Key KPIs

KPI	Value
Total Revenue	■7,328,061.58
Total Profit	■2,070,172.88
Total Orders	1,200
Average Order Value	■6,106.72
Profit Margin	28.25%

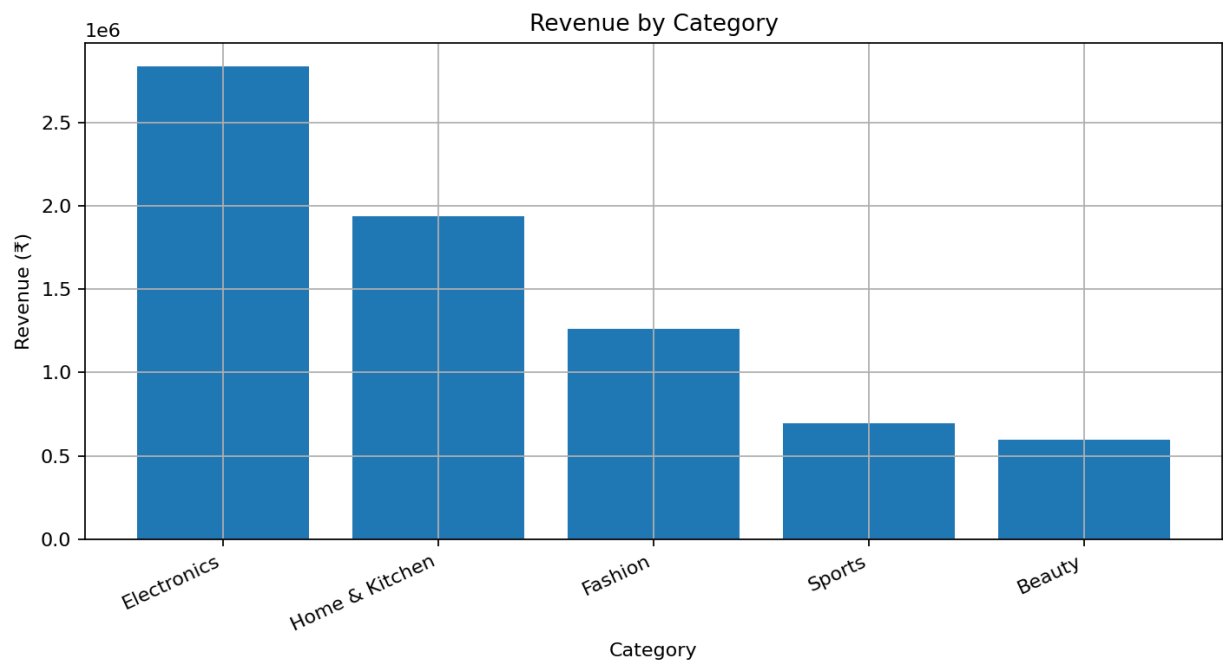
4. Sales Trend Analysis

Monthly revenue is plotted to identify changes in sales performance across the year. The trend can support planning for inventory, marketing campaigns and seasonal demand.



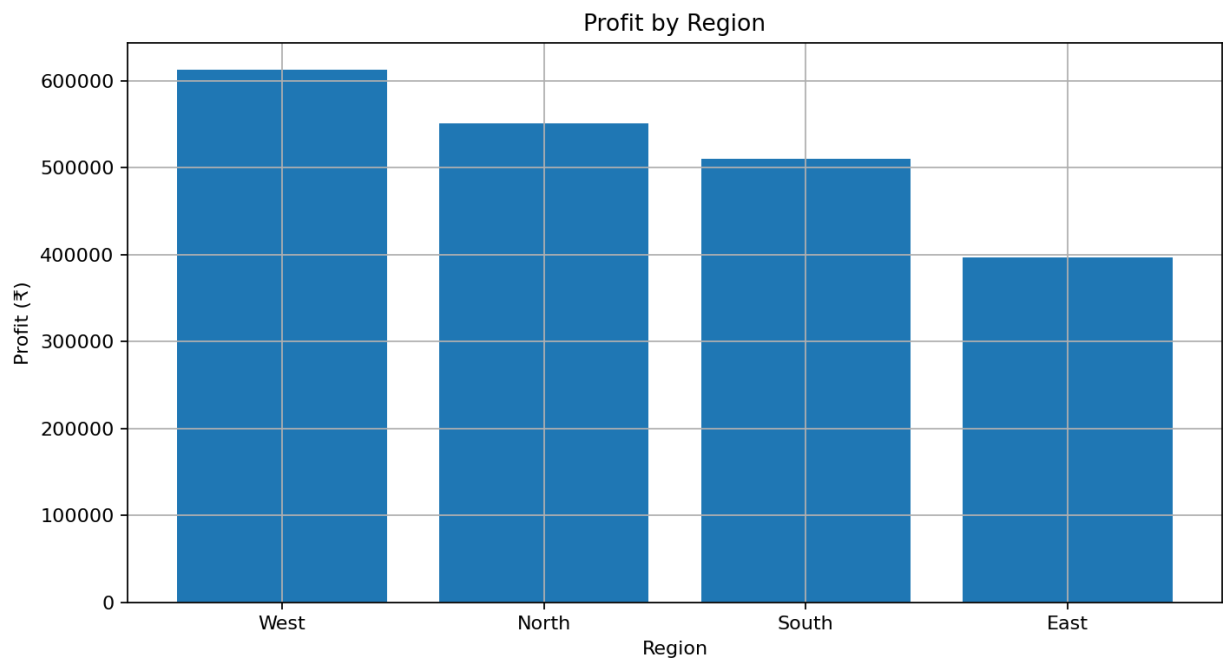
5. Category Analysis

Revenue is compared across product categories to identify categories contributing strongly to sales. Category performance can guide product assortment and promotional planning.



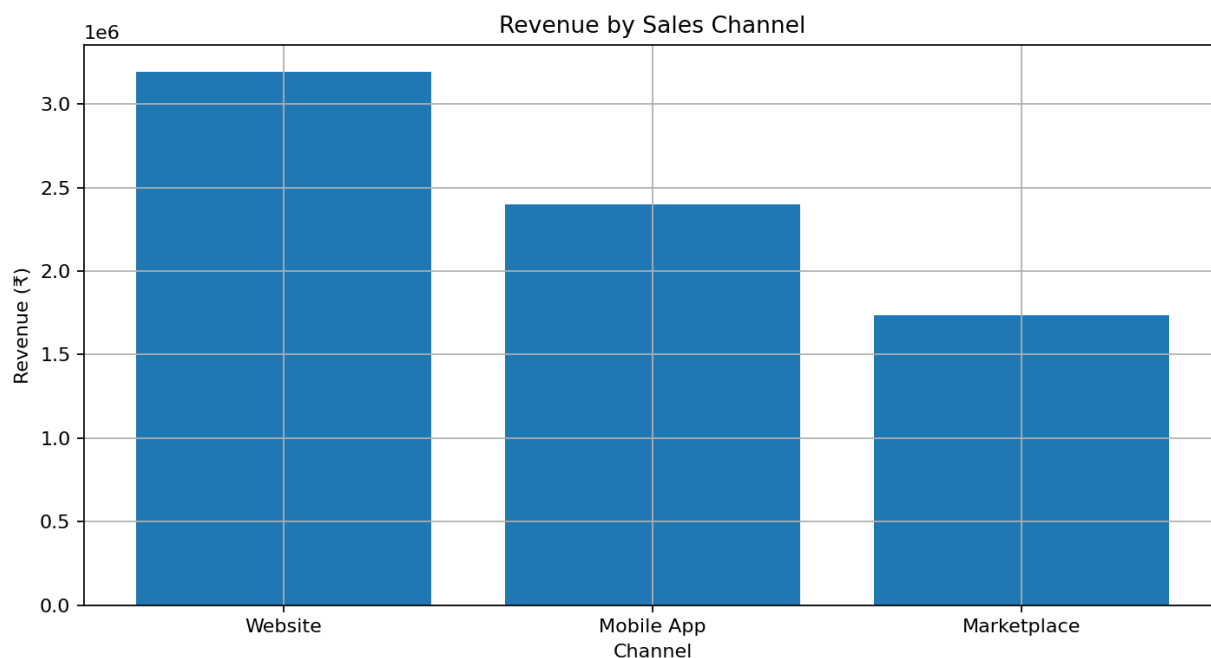
6. Regional Profitability

Regional profit is compared to understand where revenue is translating into stronger business returns.



7. Channel Analysis

Website, mobile app and marketplace revenue are compared to understand channel contribution.



8. Business Insights

- Revenue performance should be monitored month by month to identify periods requiring corrective action.
- The highest-revenue category in this project is Electronics.
- The highest-revenue region in this project is West.
- The highest-revenue channel in this project is Website.
- Profitability should be considered alongside revenue because a high-sales segment may not always produce the highest profit.

9. Recommendations

1. Track monthly revenue and investigate significant deviations. 2. Prioritize inventory and marketing around high-performing categories. 3. Review regional profit before allocating promotional budgets. 4. Compare sales channels regularly and improve the channels with weaker contribution. 5. Monitor discounting because discounts reduce realized revenue.

10. Conclusion

The project demonstrates how raw transaction-level data can be converted into KPIs, visual analysis and business recommendations. The workflow can be extended with Power BI, SQL analysis or predictive modelling in a future version.